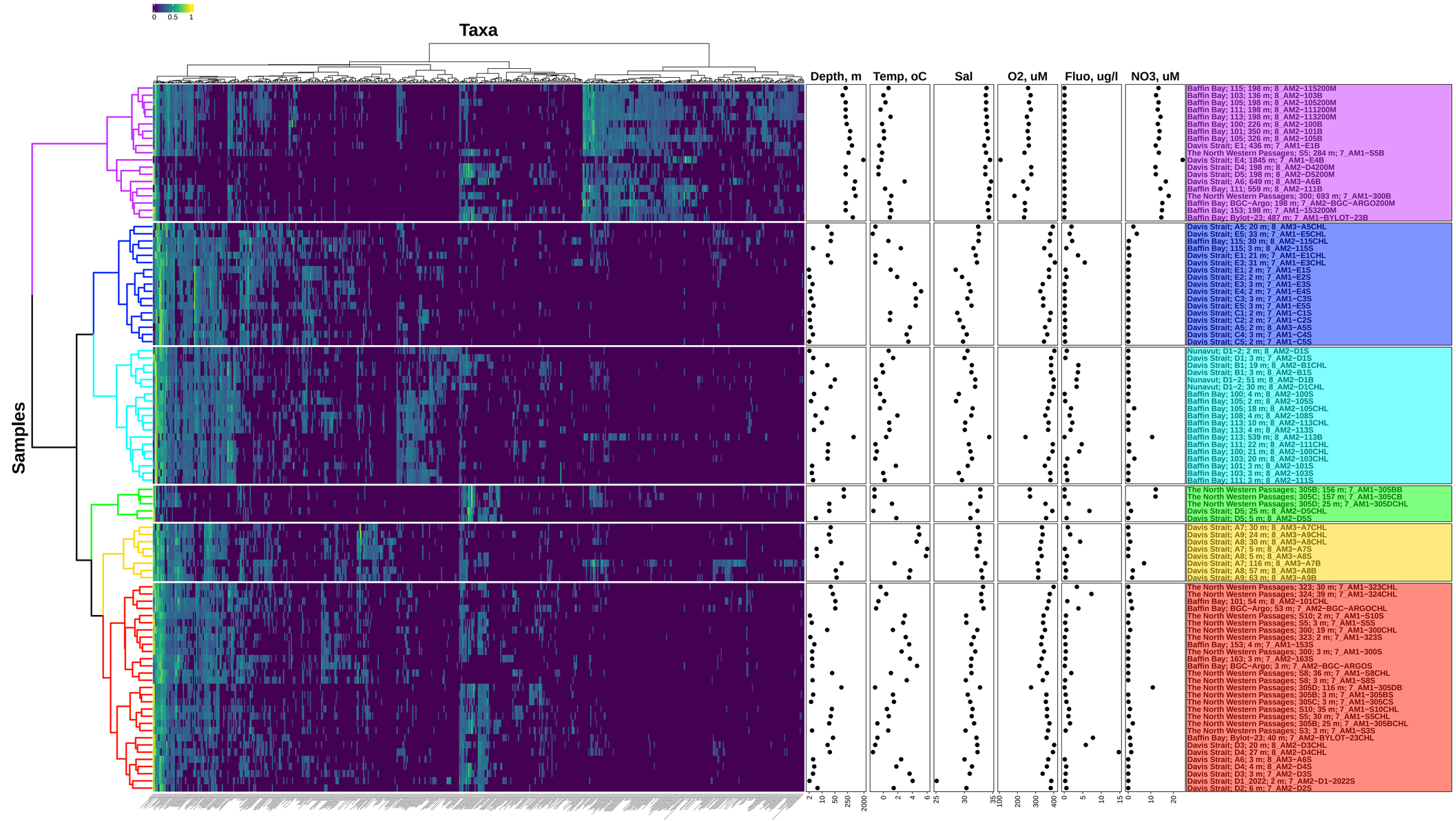
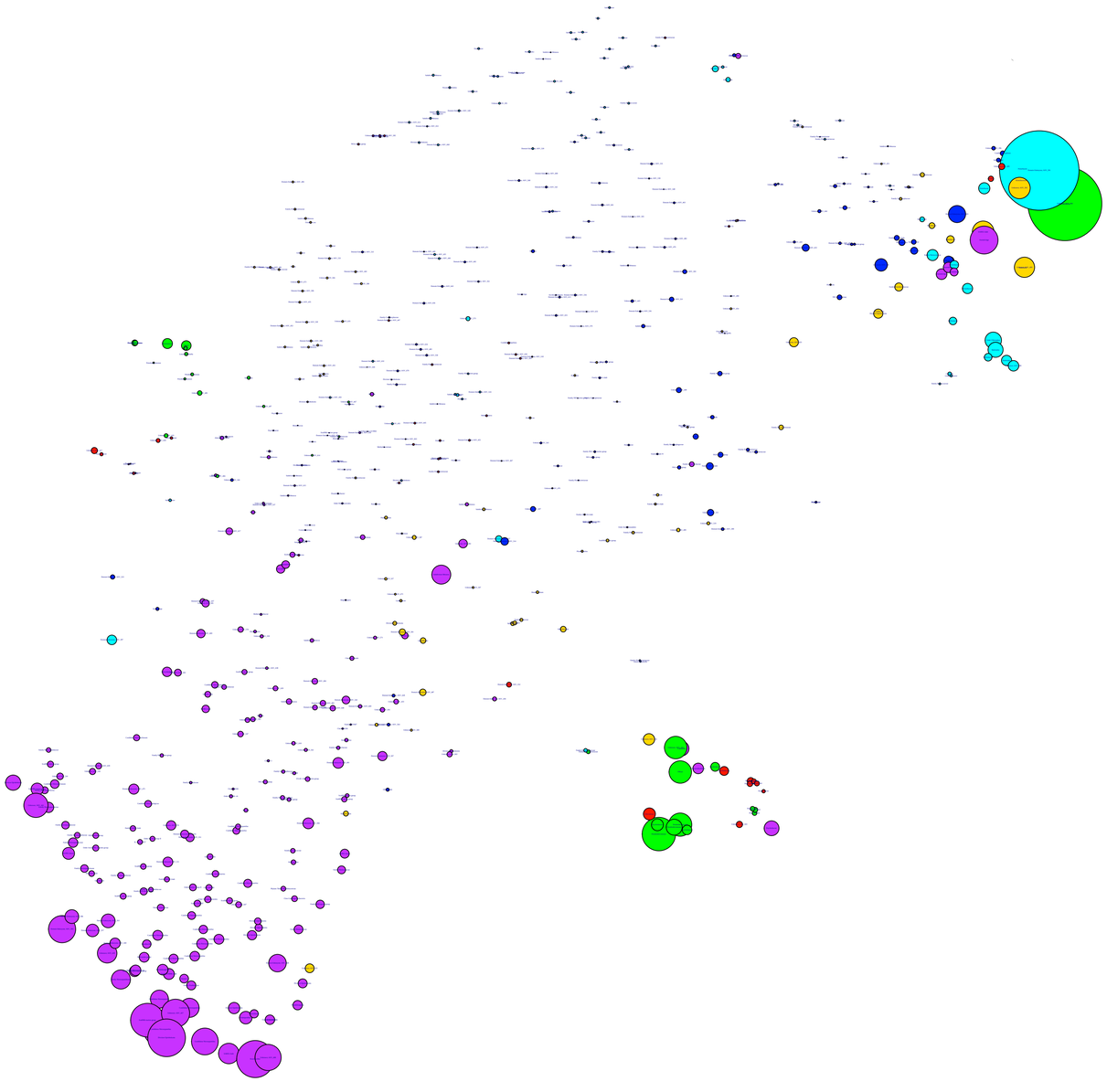
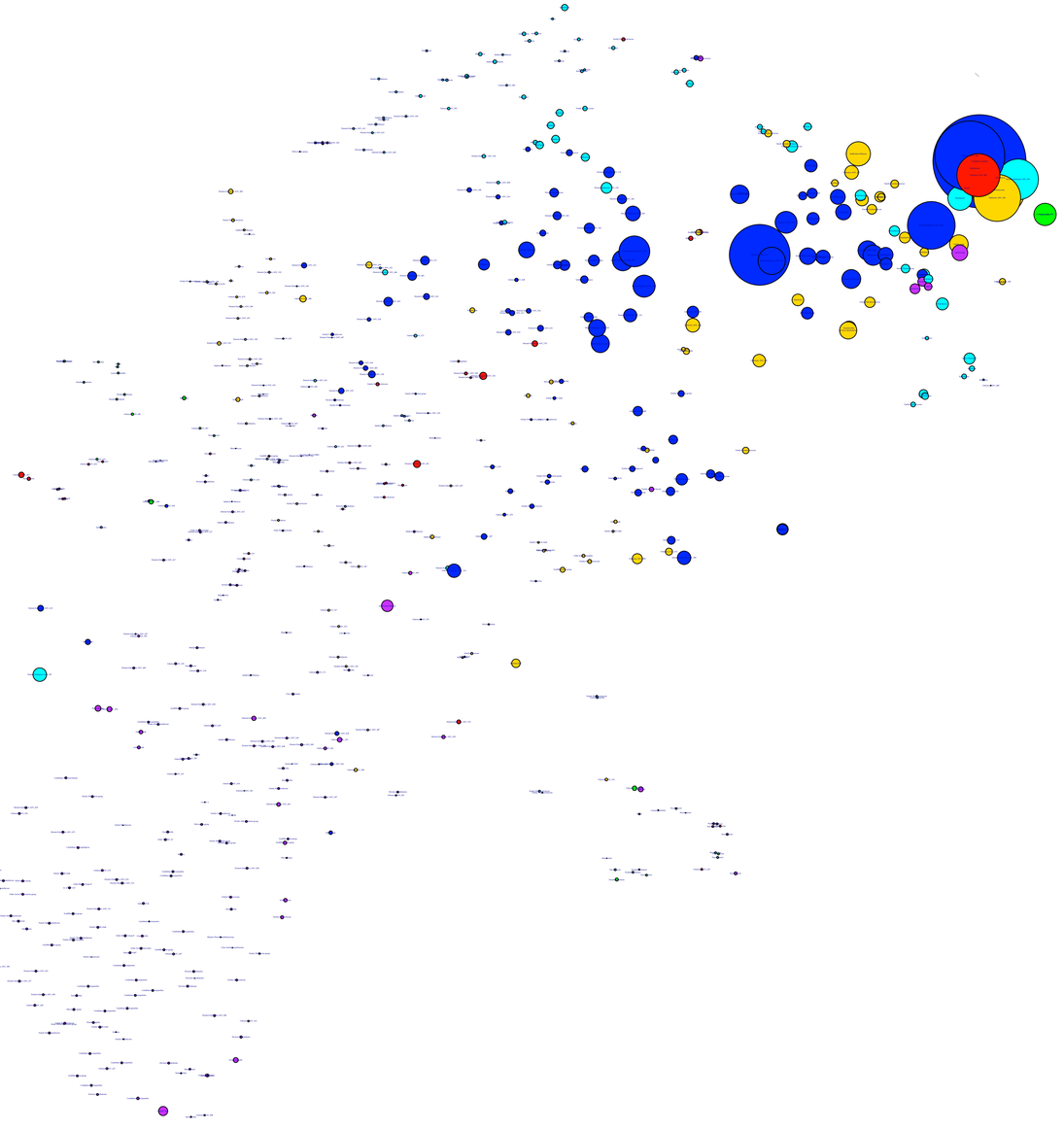
Fifth−root transformed relative abundances of microbial taxa during Leg 2 of the 2023 Amundsen expedition



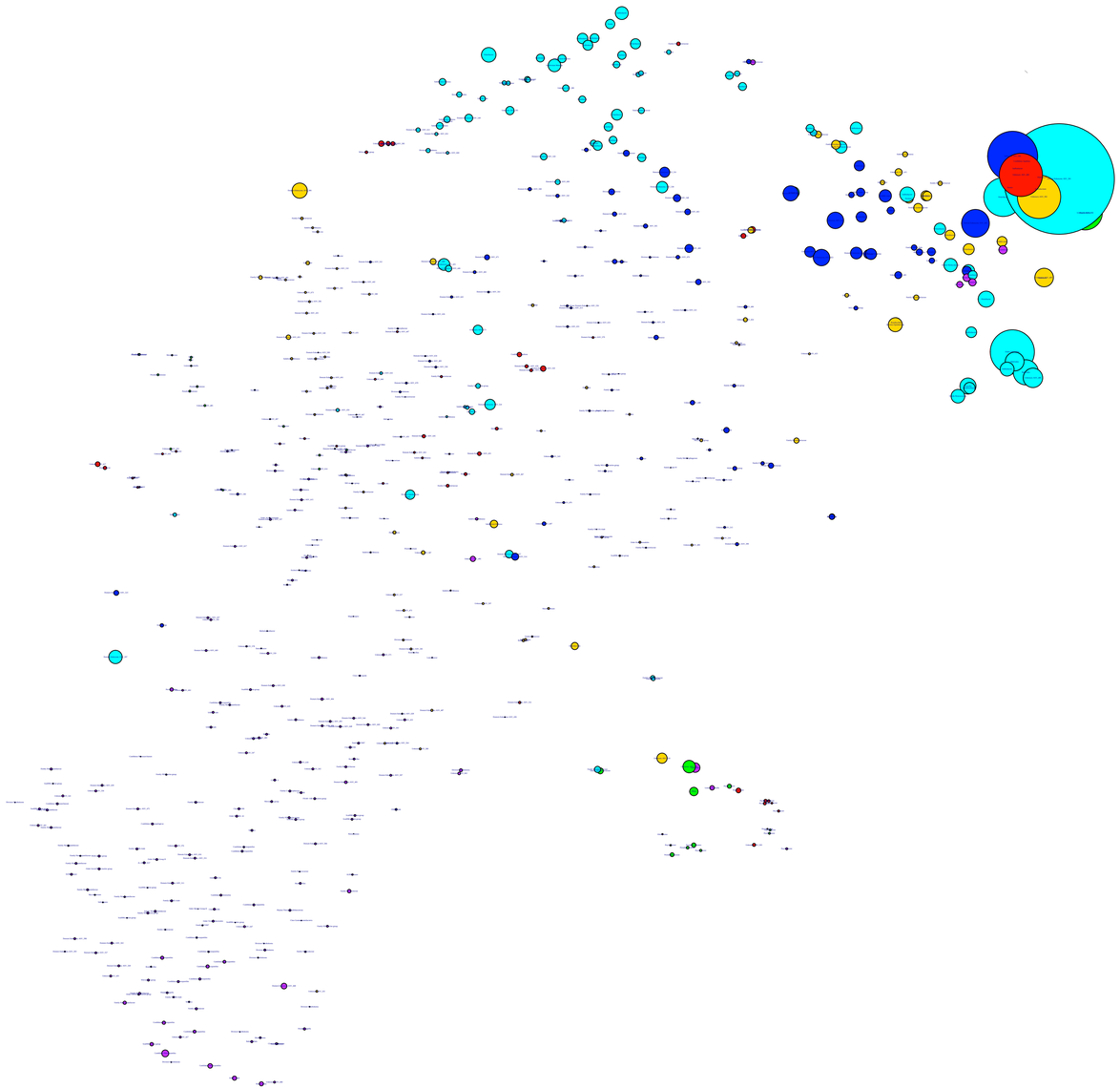
Square−root transformed relative abundances of microbial taxa in Cluster 1 (ASVs are colored by Cluster they are most associated with)



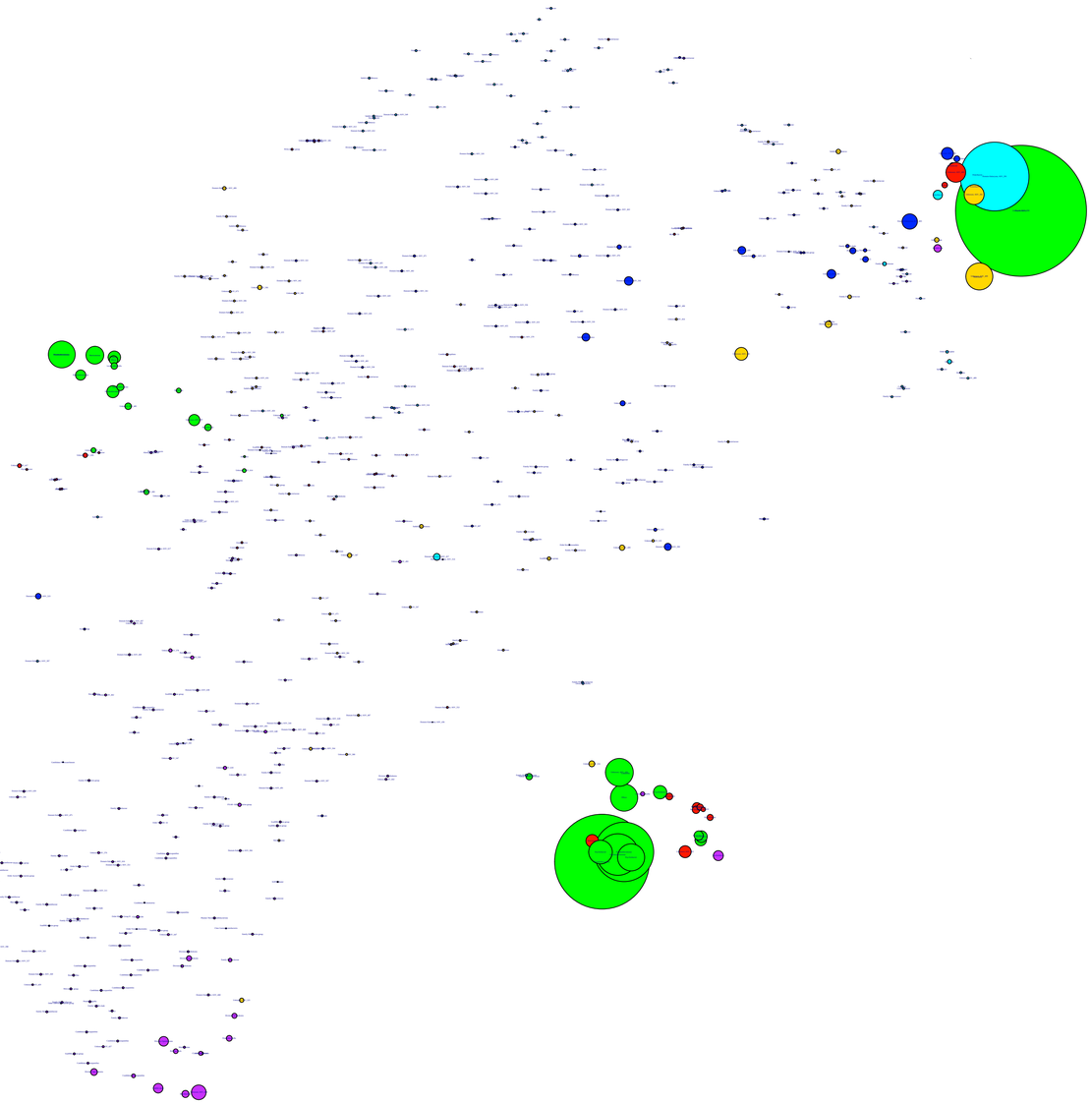
Square−root transformed relative abundances of microbial taxa in Cluster 2 (ASVs are colored by Cluster they are most associated with)



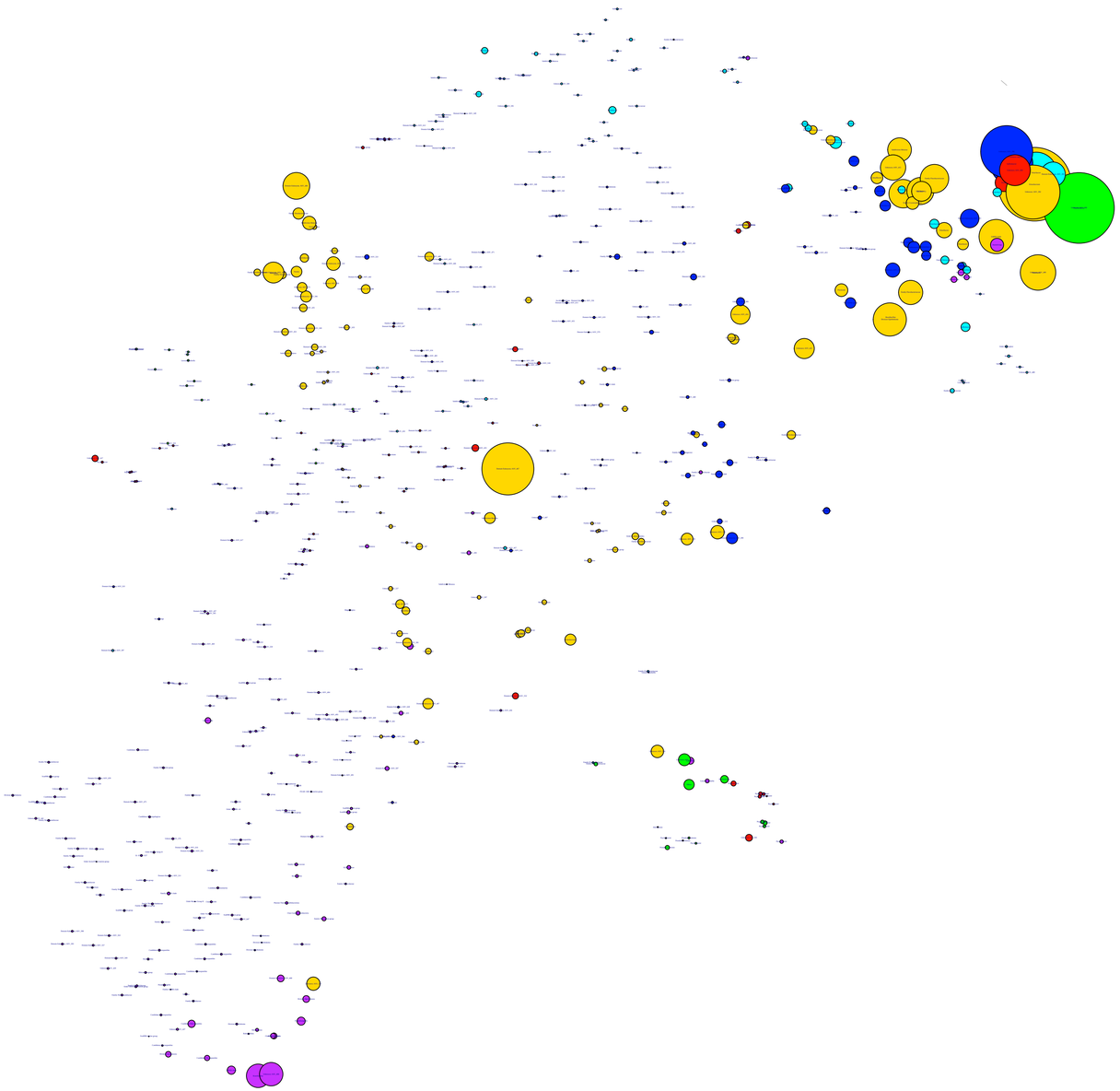
Square−root transformed relative abundances of microbial taxa in Cluster 3 (ASVs are colored by Cluster they are most associated with)



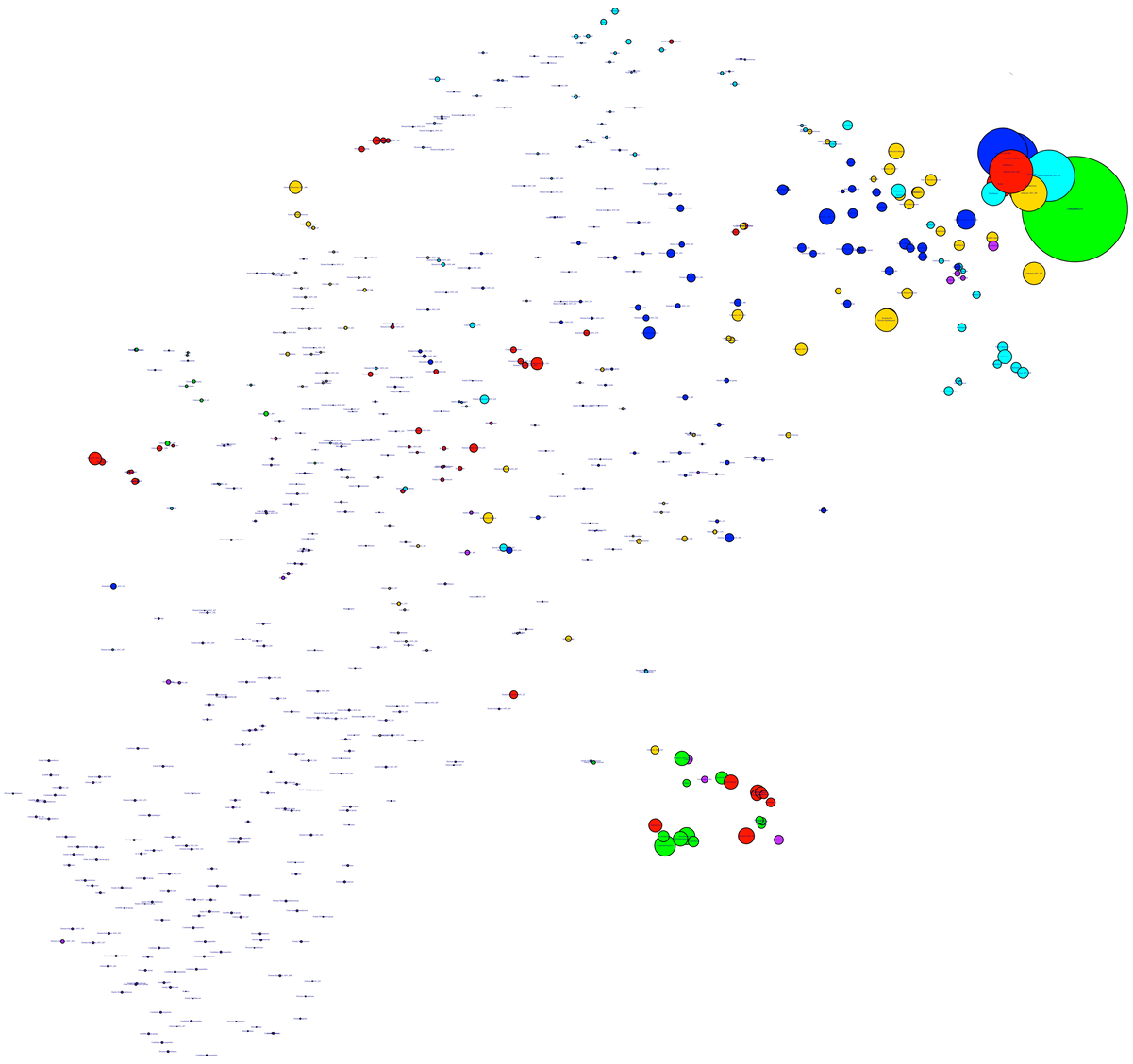
Square−root transformed relative abundances of microbial taxa in Cluster 4 (ASVs are colored by Cluster they are most associated with)



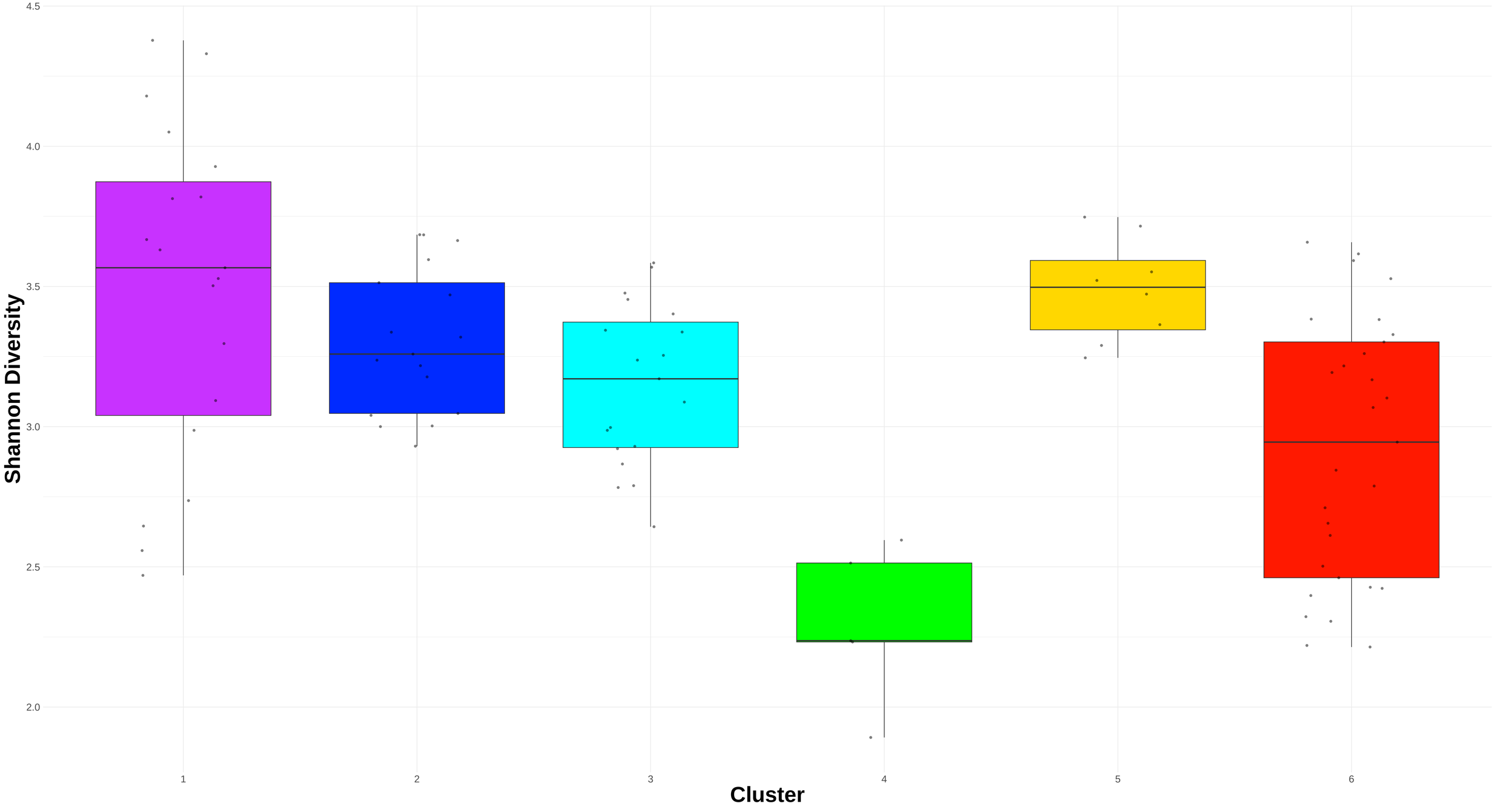
Square−root transformed relative abundances of microbial taxa in Cluster 5 (ASVs are colored by Cluster they are most associated with)



Square−root transformed relative abundances of microbial taxa in Cluster 6 (ASVs are colored by Cluster they are most associated with)



Shannon diversity by Cluster (Kruskal−Wallis test p−value = 1.63e−05)



Most abundant Cluster−specific ASVs (>1% mean relative abundance)

Cluster	ASV	Abundance
1	Opisthokonta; Division Opisthokonta; ASV_431	0.057
1	Verrucomicrobiota; Verrucomicrobiales; Roseibacillus; ASV_50	0.055
1	Thermoproteota; Nitrosopumilales; Candidatus Nitrosopumilus; ASV_32	0.048
1	Actinomycetota; Microtrichales; Sva0996 marine group; ASV_98	0.046
1	Bacteroidota; Flavobacteriales; Aurantivirga; ASV_28	0.031
1	Thermoproteota; Nitrosopumilales; Candidatus Nitrosopumilus; ASV_16	0.029
1	Pseudomonadota; Pseudomonadales; SAR92 clade; ASV_34	0.017
1	Thermoproteota; Nitrosopumilales; Candidatus Nitrosopumilus; ASV_24	0.015
1	Thermoproteota; Nitrosopumilales; Family Nitrosopumilaceae; ASV_57	0.015
1	Opisthokonta; Subdivision Metazoa; ASV_478	0.014
1	Thermoproteota; Nitrosopumilales; Candidatus Nitrosopumilus; ASV_83	0.013
2	Actinomycetota; Micrococcales; Candidatus Aquiluna; ASV_14	0.310
2	Pseudomonadota; Rhodobacterales; Sulfitobacter; ASV_13	0.094
2	Pseudomonadota; Rhodobacterales; Planktomarina; ASV_9	0.015
2	Pseudomonadota; Pseudomonadales; Family SAR86 clade; ASV_7	0.014
2	Opisthokonta; Subdivision Metazoa; ASV_422	0.012
3	Bacteroidota; Flavobacteriales; Polaribacter; ASV_5	0.280
3	Cyanobacteriota; Chloroplast; Order Chloroplast; ASV_30	0.076
3	Bacteroidota; Flavobacteriales; Polaribacter; ASV_31	0.058
3	Verrucomicrobiota; Verrucomicrobiales; Rubritalea; ASV_120	0.025
3	Verrucomicrobiota; Verrucomicrobiales; Rubritalea; ASV_181	0.014
3	Pseudomonadota; Enterobacterales; Colwellia; ASV_104	0.010
4	Pseudomonadota; Enterobacterales; Pseudoalteromonas; ASV_21	0.340
4	Pseudomonadota; Enterobacterales; Pseudoalteromonas; ASV_27	0.130
4	Pseudomonadota; Enterobacterales; Pseudoalteromonas; ASV_198	0.065
4	Bacteroidota; Flavobacteriales; Olleya; ASV_157	0.028
4	Pseudomonadota; Pseudomonadales; Psychrobacter; ASV_235	0.028
4	Pseudomonadota; Enterobacterales; Pseudoalteromonas; ASV_281	0.027
4	Pseudomonadota; Pseudomonadales; Psychrobacter; ASV_234	0.021
4	Pseudomonadota; Enterobacterales; Pseudoalteromonas; ASV_201	0.015
4	Pseudomonadota; Pseudomonadales; Marinomonas; ASV_292	0.012
5	Pseudomonadota; Pseudomonadales; Kineobactrum; ASV_3	0.210
5	Pseudomonadota; Pseudomonadales; SAR92 clade; ASV_26	0.046
5	Opisthokonta; Division Opisthokonta; ASV_394	0.043
5	Bacteroidota; Flavobacteriales; Formosa; ASV_6	0.032
5	Bacteroidota; Flavobacteriales; Family Flavobacteriaceae; ASV_46	0.032
5	Verrucomicrobiota; Verrucomicrobiales; Rubritalea; ASV_64	0.032
5	Bacteroidota; Flavobacteriales; Gilvibacter; ASV_11	0.029
5	Verrucomicrobiota; Verrucomicrobiales; Roseibacillus; ASV_15	0.024
5	Bacteroidota; Flavobacteriales; Family Flavobacteriaceae; ASV_18	0.023
5	Opisthokonta; Subdivision Metazoa; ASV_410	0.022
5	Actinomycetota; Micrococcales; ML602J−51; ASV_60	0.015
6	Pseudomonadota; Pseudomonadales; Psychrobacter; ASV_4	0.330
6	Pseudomonadota; Rhodobacterales; Yoonia; ASV_25	0.028

Cluster−specific values of defining environmental parameters

Cluster	Location_freq	Bottom depth, m	Target depth	Depth, m	Depth_cor, m	Temp, oC	Sal	Fluo, ug/l	Bacteria, 1/ml	Eukaryota, 1/ml	NO3, uM	NO2, uM
1	Baffin Bay	794	bottom	136 − 1845	198 − 693	0.34	34.0	0.07	79567	262	14.46	0.04
2	Davis Strait	1053	surf	2 − 33	2 − 31	1.87	30.7	1.06	532784	5412	0.42	0.02
3	Baffin Bay	318	surf	2 − 539	2 − 51	0.09	30.7	1.81	182456	3342	0.93	0.02
4	The North Western Passages	826	bottom	5 − 157	25 − 156	−0.17	32.0	1.78	740966	5310	5.12	0.03
5	Davis Strait	90	bottom	5 − 116	5 − 63	4.37	32.7	1.10	354796	1631	1.48	0.10
6	The North Western Passages	746	surf	2 − 116	2 − 54	1.37	31.3	1.94	593843	6317	0.74	0.02